

**A commentary to the
European Concerted Action
“Surface Electromyography for Non Invasive Assessment of Muscles”
(SENIAM, 1996-1999)
Twenty years later
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Roberto Merletti, Ph.D
LISiN, Politecnico di Torino, Italy,
SENIAM Management Group**

1. Objectives:

The European Concerted Action “Surface electromyography for non invasive assessment of muscles “ (SENIAM) was carried out with EU support in 1996-1999 with two main objectives:

1. To exchange knowledge and experiences among different countries, disciplines and professions, and to enhance transfer of knowledge from research labs to teaching and clinical institutions.
2. To develop recommendations on key items that were limiting exchange of knowledge and experience with respect to clinical surface electromyography (sEMG) data and findings.

2. Project Management Group

The SENIAM project was co-ordinated and managed by:

Dr. H.J. Hermens and B Freriks – *Enschede - the Netherlands*,
Prof. R. Merletti - *Torino - Italy*
Prof. Dr. G. Rau and Dr. C. Disselhorst-Klug - *Aachen - Germany*
Prof. Dr. D. F. Stegeman - *Nijmegen - the Netherlands*
Dr. G.M. Hägg - *Solna - Sweden*

Researchers and clinicians of nine countries participated to the project:
Belgium, Finland, France, Greece, Italy, Germany, the Netherlands, Sweden, Switzerland

3. Products and Results of the Project

The Project supported the exchange of researchers among the participating countries and produced eight deliverables/books collected in a CD-ROM and in the website www.seniam.org. Volume 8 is a summary of recommendations and is available in this website, with permission. These books are:

SENIAM 1. European Activities on Surface Electromyography.
Proceedings of the first general SENIAM workshop, Torino, Italy, September 1996
Hermie J. Hermens, Roberto Merletti, Bart Freriks

SENIAM 2. European Applications of Surface Electromyography.
 Proceedings of the second general SENIAM workshop, Stockholm, Sweden, June 1997
 Hermie J. Hermens, Göran Hägg, Bart Freriks

SENIAM 3. Surface Electromyography Application Areas and Parameters.
 Proceedings of the third general SENIAM workshop, Aachen, Germany, May 1998
 Hermie J. Hermens, Günter Rau, Catherine Disselhorst-Klug, Bart Freriks

SENIAM 4 Future Applications of Surface Electromyography
 Proceedings of the fourth general SENIAM workshop, Hertogenbosch, the Netherlands,
 September 1999
 Hermie J. Hermens, Bart Freriks

SENIAM 5. The State of the Art on Sensors and Sensor Placement Procedures for Surface
 Electromyography: A proposal for sensor placement procedures. 1997
 Hermie J. Hermens, Bart Freriks

SENIAM 6. State of the Art on Modelling Methods for Surface Electromyography, 1998
 Hermie J. Hermens, Dick Stegeman, Joleen Blok, Bart Freriks

SENIAM 7. The State of the Art on Signal Processing Methods for Surface
 Electromyography. 1999
 Hermie J. Hermens, Roberto Merletti, Herve Rix, Bart Freriks

SENIAM 8. European Recommendations for Surface Electromyography. Results of the
 SENIAM project . 1999
 Hermie J. Hermens, Bart Freriks, Roberto Merletti, Dick Stegeman, Joleen Blok, Günter Rau,
 Catherine Disselhorst-Klug, Göran Hägg

The tables of content of these books are available on www.seniam.org.

Volume 8 has been translated in Italian and is available from the Italian version of this
 website.

4. Twenty years later

4.1 Strengths of the SENIAM Project

The SENIAM Project came 15 years after the fundamental textbook by J. Basmajian and C. J. De Luca “Muscles alive: their functions revealed by electromyography” Williams & Wilkins, USA, Fifth Edition, 1985, with the purpose of disseminating and systematizing the growing sEMG knowledge and methodology. The SENIAM Project provided recommendations and guidelines concerning the use of surface EMG in the research and clinical worlds in particular in the fields of neuromuscular physiology, sport, prevention and occupational medicine. Most papers published in these fields in the last two decades refer to SENIAM Guidelines in their methodology sections.

Many subsequent initiatives were directly or indirectly inspired by the SENIAM Project. Thousands of papers in the last two decades further investigated and illustrated the nature and the information content of the surface EMG providing new knowledge and material that updated and tuned the original recommendations. A few textbooks (listed in this website) have

been written to disseminate this knowledge among students in the many fields involving sEMG. Fig. 1 provides a general perception. See section 4.3.

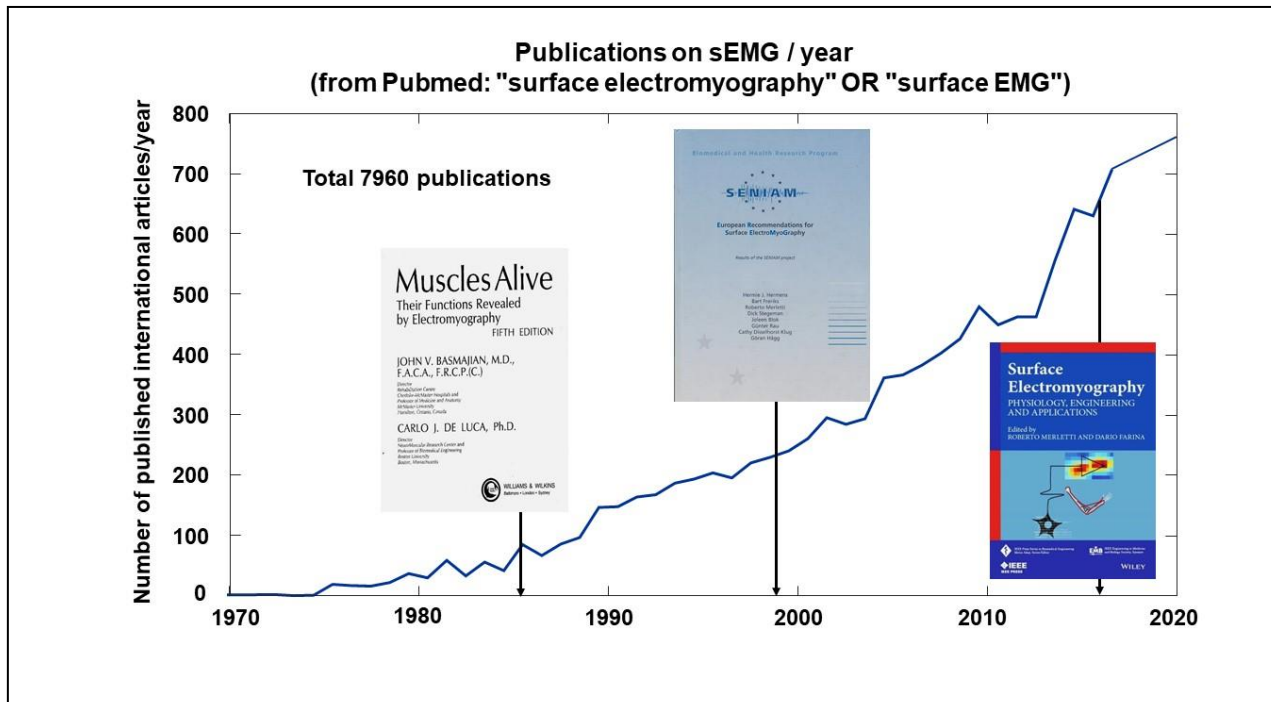


Fig. 1. Plot of the sEMG publications per year resulting from Pubmed using the search string < "surface electromyography" OR "surface EMG"> in February 2021. Most of the articles are written by engineers. The total number of publications drops to 51 if < AND "clinical applications"> is added to the search string.

4.2 Weaknesses of the SENIAM Project

Despite the efforts of the SENIAM partners, the SENIAM recommendations never generated a set of accepted standards and protocols to be followed in clinical practice. Despite pressure by researchers, time was not mature for a substantial participation of clinicians in a project that many of them considered of engineering pertinence. The work has been well received by scholars and researchers but its consideration by teachers and schools in the fields of physiotherapy and occupational therapy has been very country-dependent. This fact limited further projects and progress in the educational field, but certainly not in the research fields, resulting in a widening gap between available knowledge and its clinical applications. In some countries this gap is dramatic and has led to vicious circles such as indicated in Fig. 2. Interventions are needed to correct this situation and break the circle. The creation of graduate programs with strong interdisciplinary connotation is one of such interventions.

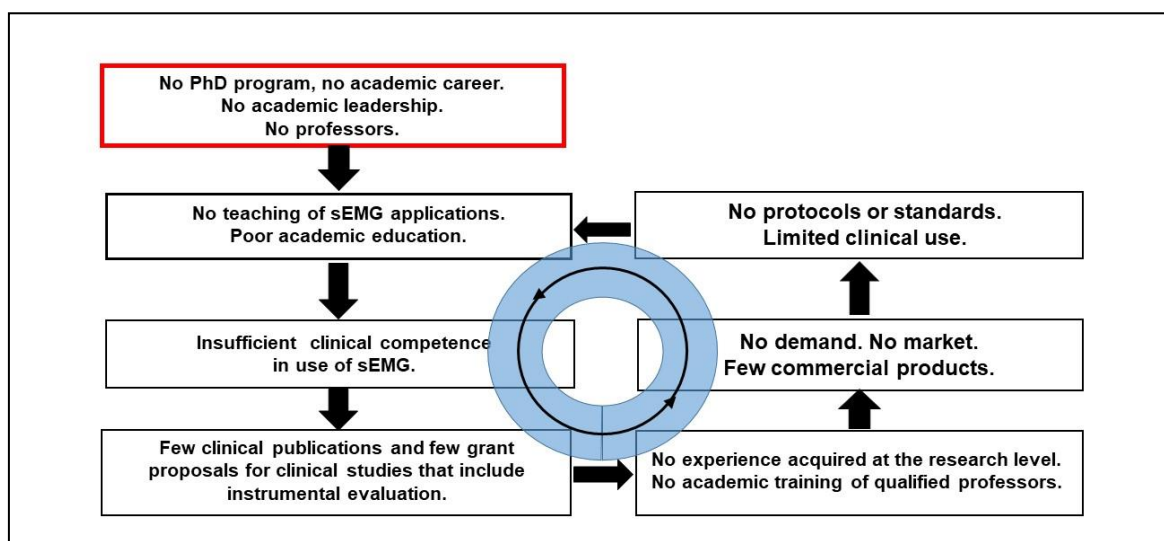


Fig. 2. A vicious circle developed in countries who do not offer Ph.D. or post graduate education in the fields of physiotherapy and occupational therapy. The situation is less critical in the fields of movement sciences and kinesiology. Improving the level and quality of teaching by introducing a higher level of education, including technological knowledge, is one way to break the circle. An alternative way is to introduce new figures with a predominant technical knowledge.

4.3. Post-SENIAM developments: the last twenty years

Extensive new sEMG knowledge and technologies have been developed since the SENIAM Project and are described in a number of books. Some of the recommendations need updating and adaptation to such new knowledge, in particular concerning the development of 2D High Density electrode technology (electrode size and distance, signal processing and clinical interpretation) and the technology for the decomposition of sEMG into the constituent trains of motor unit action potentials. Remarkable differences still exist between EU countries and require additional EU (as well as worldwide) efforts for standardization of the technique, definition of clinical protocols and teaching and training of new technical and/or clinical operators able to address the multidisciplinary nature of the field.

Conditions limiting the dissemination, teaching and clinical use of sEMG (as well as of other measurement techniques) still exist in many countries where technology seems to be more welcome when it provides “therapeutic” tools than when provides sensors and measurement tools. An authoritative analysis of the problem is provided by the Editorial Project: “Surface Electromyography: barriers limiting widespread use of sEMG in clinical assessment and neurorehabilitation” sponsored by the open-access journal Frontiers in Neurology/Neurorehabilitation (<https://www.frontiersin.org/research-topics/11157>). The Project is described in a free e-book with 18 contributions from seven countries and 80 authors (33 engineers, 16 MDs, 18 physio-therapists and occupational therapists, 13 movement scientists) who are highly respected experts in the many fields of sEMG. The Project received about 30.000 views (February 2021), indicating the high interest in the subject. Other efforts are carried out by the Journal of Electromyography and Kinesiology with the publication of open access tutorials and consensus papers.

4.4. Post-SENIAM developments: the next twenty years

The sEMG technology is rapidly growing and major advances are expected in the near future. Some will focus on the acquisition of new knowledge but many will focus on the solving of technical problems that make the clinical applications relatively difficult or cumbersome.

Wearable elastic sleeves containing hundreds of electrodes are becoming available in research labs and software for the automatic selection of relevant channels and for the interpretation of the associated information is being developed. Without proper user education the gap between sophisticated technology and user clinical experience may widen. Inclusion of “intelligence” in new devices, to facilitate signal interpretation, is not necessarily a good solution, although proposed by some clinicians as a possible approach.

As happened in other interdisciplinary fields, the training of new hybrid figures may be an alternative to increasing the knowledge and competence of the existing ones that are already overloaded. The new figure of the “clinical technologist” is the option described in

<https://www.tudelft.nl/en/2020/3me/june/clinical-technologists-officially-registered-healthcare-professionals/> (March 2021).

Its area of expertise includes a) the correct application of existing measurement procedures, b) the design and development of new methods and therapies using technology, and c) the carrying out of complex technical procedures in the field in which the clinical technologist is trained.

While other options may be possible, international collaboration and EU support/coordination are badly needed to smooth out the differences between countries in the field of technology education in the health delivery sector, and specifically in the sEMG field. This requires a greater involvement of the clinical operators in achieving higher levels of education and in writing international proposals to address the issue of technology management.

The international and national scientific societies should take a role in proposing a new SENIAM Project focused on technology transfer, user education and training of new figures with interdisciplinary knowledge.